

SESSION 4

Margin Trading, Leverage & Short Selling

How borrowed money changes investor behaviour — and destroys wealth

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Leverage — Friend or Enemy? The Appeal of Borrowed Money

You have ₹1 lakh. You believe Reliance will rise 20% in three months. Would you borrow ₹4 lakh more to amplify your profit?

✓ IF YOU'RE RIGHT (+20%)

Own capital:	₹1,00,000
Borrowed funds:	₹4,00,000
Total position:	₹5,00,000
Gain (20%):	₹1,00,000
Borrowing cost (8%):	-₹8,000
Net profit:	₹92,000
Return on own capital:	92% 🚀

✗ IF YOU'RE WRONG (-20%)

Own capital:	₹1,00,000
Borrowed funds:	₹4,00,000
Total position:	₹5,00,000
Loss (20%):	-₹1,00,000
Borrowing cost (8%):	-₹8,000
Net loss:	-₹1,08,000
Return on own capital:	-108% 💀



Core Insight: Leverage doesn't change the probability of being right — it magnifies the consequences of being wrong.

How Margin Buying Works: The Mechanics of Borrowed Conviction



Investment
Conviction



Margin
Account



Borrowed
Funds



Leveraged
Position



Wealth Creation
or Ruin

Initial Margin (SEBI India)

SEBI mandates minimum margin requirements for equity futures. For equity intraday, brokers may allow up to 5x leverage (20% margin). F&O requires SPAN + Exposure margins as defined by exchanges.

📌 *Nifty Futures margin: ~₹1.0–1.2 lakh per lot for a ₹10 lakh exposure. That's ~10–12% margin = ~8x leverage on notional value.*

🔗 sebi.gov.in/margin-framework

Leverage Amplification Math

With 5x leverage: a 10% favourable move = 50% gain. A 20% adverse move = 100% loss of capital. A 25% adverse move = losing 25% MORE than you started with — you owe the broker.

📌 *Leveraged ETF reality: Direxion 3x Bull S&P 500 ETF (SPXL) fell 80% in the 2020 COVID crash — vs S&P 500's 34% decline. Volatility drag compounds losses.*

🔗 direxioninvestments.com/products/spxl

Who Uses Margin Legally?

Hedge funds for short-term arbitrage and pair trades. Proprietary trading desks for intraday capture. Real estate investors (mortgages = leverage). Long-term investors — rarely, and only for specific structures.

📌 *Warren Buffett on leverage: 'We have never used leverage and never will. Smart people go broke by combining a good business idea with too much debt.'*

🔗 berkshirehathaway.com/letters/2010ltr.pdf

Margin Requirements: Who Really Controls Your Leveraged Position?



INITIAL / PERCENTAGE MARGIN

What it is

The minimum deposit required to open a leveraged position — expressed as a % of total position value.

Who sets it

SEBI (India) sets minimum floors. Individual brokers may impose higher requirements based on volatility and client risk profile.

SEBI India norms

Equity delivery: no mandated leverage cap. Intraday equity: brokers may allow up to 5x. F&O: SPAN margin + Exposure margin (dynamic).

Investor mistake

Treating margin requirement as a safety guarantee. The margin only protects the broker — not the investor — against adverse moves.

sebi.gov.in/margin-framework | nseindia.com/products/fo-margins



MAINTENANCE MARGIN & MARGIN CALL

1

Market Falls

Position value drops. Equity in account shrinks.

2

Equity < Maintenance Level

Account equity falls below broker's minimum threshold (e.g., 25–30% of position).

3

Margin Call Issued

Broker demands additional funds within hours (sometimes minutes during crashes).

4

Investor Cannot Pay

Cash not available, other assets illiquid, or market moving too fast.

5

Forced Liquidation

Broker sells positions at market — often at the worst possible moment. Loss is locked in permanently.



Key Insight: The investor believes they control risk. The margin call proves the broker controls the position.

Short Selling: Profiting From Falling Prices

THE MECHANICS OF A SHORT SALE

1. Borrow

Short-seller borrows shares from a broker/lender (paying a borrowing fee)

2. Sell

Immediately sells borrowed shares in the open market at current market price

3. Wait

Waits for price to decline (if correct) — carries borrowing costs + unlimited upside risk

4. Buy Back

Buys the same shares back at the (hopefully lower) market price — called 'covering the short'

5. Return & Profit

Returns shares to lender.
 $\text{Profit} = \text{Sale price} - \text{Repurchase price} - \text{Borrowing fees}$

✓ WHY SHORT SELLING IS ESSENTIAL

Price discovery

Short sellers do fundamental analysis to identify overvalued stocks. They publish research that corrects mispricing before it becomes catastrophic.

Fraud detection

Enron, Wirecard, Satyam — all identified by short-sellers before regulators. Short sellers have a financial incentive to find fraud.

Market liquidity

Short sellers provide liquidity by being willing sellers when others are buying. They improve market depth and reduce spreads.

Checks on mania

During bubble periods, short sellers provide the counterforce against irrational euphoria — even if they are early and get punished for it.

⚠ RISKS & CONTROVERSIES OF SHORTING

Unlimited losses

A long investor can lose 100% of investment. A short seller can lose INFINITELY more — stocks have no upper price limit.

Short squeezes

If price rises, short sellers must cover at a loss. Rising price forces more covering = further price rise = GameStop-style spiral.

Timing difficulty

Being right about overvaluation is not enough. Timing the correction is extremely difficult. 'Markets can remain irrational longer than you stay solvent.'

Regulatory risk

Regulators often ban short selling during crises. India: SEBI banned short selling in 2008 crisis. This creates forced covering at worst moments.

Adani–Hindenburg (2023): When Short-Seller Research Moves Markets

TIMELINE: January–March 2023

Jan 24

Hindenburg Research publishes 100-page report alleging accounting fraud, stock manipulation, and governance failures at Adani Group

Jan 25–27

Adani stocks collectively lose ₹5+ lakh crore in market cap across 3 trading sessions. Adani Enterprises falls 28% in 2 days

Jan 28–29

FPO proceeds of ₹20,000 crore returned to investors. Adani scraps FPO to 'insulate investors from potential losses'

Feb–Mar

Supreme Court panel investigation ordered by SEBI. Further ₹6+ lakh crore cumulative decline across Adani group companies

Mar–Jun

GQG Partners invests \$1.87B — partial recovery begins. Information eventually 'priced in' — classic market efficiency in action

 **Information Power:** A 100-page research report triggered a ₹12 lakh crore market cap loss. This is price discovery working — rapidly.

 **Activism vs Manipulation?:** Is Hindenburg a legitimate short-seller doing research, or a manipulator profiting from market panic? The debate remains open.

 **Markets Recover:** Stocks recovered substantially over 12–18 months. Information gets priced in — eventually. Timing is the real challenge.

GameStop (2021): When Retail Investors Crushed a Short Squeeze

2,400%

GME share price rise in 3 weeks | \$5B+ losses for hedge funds | Melvin Capital required \$2.75B emergency bailout



WHAT HAPPENED

- GameStop was heavily shorted (140%+ float shorted — more shares shorted than existed)
- Reddit community r/WallStreetBets (5M users) coordinated to buy GME and GME call options
- As price rose, short-sellers forced to 'cover' (buy back shares) — driving price higher still
- Classic 'short squeeze': shorts' losses fed the rally. Price peaked \$483 (from \$17 in Jan 2021)
- Robinhood restricted buying on Jan 28 — controversially — citing clearinghouse margin calls
- SEC investigation followed. Systemic questions about clearing infrastructure exposed.

 sec.gov/gamestop-meme-stock-report



WHAT MARKETS LEARNED

Short selling has unlimited upside risk

If 140% of float is shorted and the stock rises, every short-seller faces growing losses with no ceiling.

Social media can move markets

Reddit coordination created a feedback loop that outmaneuvered institutional risk models temporarily.

Liquidity ≠ safety in crisis

Robinhood's clearinghouse needed \$3B+ overnight — platform had to restrict retail trading to meet obligations.

Markets can remain irrational

Keynes: 'Markets can remain irrational longer than you can remain solvent.' Melvin Capital proved this again.

Timing short positions is brutal

Even if GME was overvalued at \$100, shorting at \$100 and covering at \$483 is catastrophic — being early = being wrong.

 sec.gov/meme-stock-report-2021



GameStop Lesson: Short selling has unlimited downside risk. Markets can remain irrational — longer than your capital can survive.

Archegos Capital (2021): How \$20 Billion Disappeared in 2 Days

\$20B+

Peak AUM

5–8x

Total Return Swap Leverage

\$10B+Bank Losses (Credit Suisse,
Nomura)**2 Days**

Time to Complete Collapse



WHAT HAPPENED — AND WHY IT SHOCKED

- Bill Hwang's family office used Total Return Swaps (TRS) — synthetic leverage hidden from public disclosure
- TRS allowed Archegos to hold 5–8x leveraged positions in concentrated stocks (ViacomCBS, Discovery, Baidu)
- No SEBI/SEC disclosure required for swaps — regulators & banks had no visibility into total exposure
- ViacomCBS stock fell 30%+ in a week. Archegos failed to meet margin calls from multiple banks simultaneously
- Banks (Goldman Sachs, Morgan Stanley) raced to liquidate first. Credit Suisse moved too slowly — lost \$5.5B
- Systemic risk: same concentrated positions across multiple prime brokers — all unaware of each other



REGULATORY & MARKET LESSONS

Leverage can be hidden

Total Return Swaps don't require 13F SEC disclosures — regulators had no view into Archegos's real exposures.

Concentration kills

Holding >50% of float in ViacomCBS through swaps meant any sell-off created catastrophic illiquidity.

Maintenance margin is the kill switch

The moment equity < maintenance level, brokers act autonomously — the investor has no say in timing or price.

Speed of collapse surprises everyone

\$20B of value gone in 48 hours. Not a fraud — pure leverage + concentration + forced selling cascade.

SEBI tightened India rules post-Archegos

SEBI updated prime broking and Total Return Swap disclosure norms for India — citing Archegos as the trigger.



Archegos Lesson: The greatest risk is being forced to act when you do not want to act. Leverage removes that choice.

Long-Term Capital Management:

25x

Balance Sheet Leverage (125x on some positions) | \$4.6B loss in 4 months (1998) | Fed-organised \$3.6B bailout to prevent systemic collapse

Who Was LTCM?

Founded 1994 by John Meriwether (Salomon Brothers bond legend), 2 Nobel Prize economists (Merton, Scholes), former Fed vice chair. AUM: \$4.72B. Leverage: 25–125x. Considered the smartest fund in the world.

 federalreserve.gov/ltcn-bailout

What Went Wrong

LTCM's models assumed normal market correlations. Russia defaulted on debt (Aug 1998). Global 'flight to quality' made ALL asset classes correlate to 1. Every position moved against them simultaneously. Models had never seen this regime.

 wsj.com/ltcn-failure

The Rescue

Federal Reserve Bank of NY organized \$3.6B rescue by 14 major banks. Not a government bailout — but Fed-orchestrated. LTCM had \$125B in assets, \$1.25 trillion in derivatives. Failure would have cascaded globally.

 newyorkfed.org/ltcn

The Timeless Lesson

Intelligence cannot eliminate risk. Models assume history repeats. Crises occur precisely when assumptions fail. The rarer the event your model hasn't seen, the more lethal it is when it finally arrives with leverage.

 berkshirehathaway.com/letters/1998ltr.pdf



LTCM Lesson: Intelligence cannot eliminate risk. Models fail precisely when leverage is highest and losses are most catastrophic.

The Psychology of Leverage: Why Smart Investors Repeat the Same Mistakes

Overconfidence Bias

Bull markets create the illusion that winning trades require skill — not favourable conditions. Leverage amplifies gains in bull markets, making investors believe they have edge they don't.

📌 *NSE/SEBI data (2023): 89% of F&O traders lost money over 3 years. Most had profitable first month — then added leverage. Survivorship bias hides the losses.*

🔗 sebi.gov.in/fo-study-2023

Availability Bias

Investors remember the spectacular winners — Archegos before collapse, LTCM at peak returns. They do not dwell on the survivors' counterparties. The wins are visible; the wipeouts aren't.

📌 *Indian retail trading apps (Zerodha, Upstox) show P&L dashboards prominently. Profits feel larger with leverage. Losses come suddenly — margin calls have no warning.*

🔗 zerodha.com/varsity/module/behaviour

Gambler's Fallacy

After a leveraged loss, investors believe the next trade 'must' win to recover. They increase leverage to recover faster — the classic path to ruin. Loss recovery requires geometrically more gain.

📌 *Retail F&O traders who lose ₹1 lakh, double up to recover. A 50% loss requires a 100% gain to break even. Leverage makes this gap geometrically larger.*

🔗 nseindia.com/research/derivatives-data

The Invincibility of the Bull

In bull markets, every leveraged bet seems justified by rising prices. Leverage appears to be 'working'. Margin requirements seem academic. The crash that reveals the fragility is always unexpected.

📌 *India bull market 2021–2024: Nifty went from 8,000 to 25,000. Many retail traders added leverage throughout, believing the trend was permanent. 2024 correction exposed them.*

🔗 nseindia.com/market-data/historical

🔑 **Behavioural Insight:** Leverage doesn't create bad decisions — it makes the consequences of existing bad decisions catastrophic and irreversible.

India's Retail Derivatives Boom: Access Without Expertise

89%

Retail F&O traders lose money (SEBI 2023)

₹50,000cr

Daily NSE F&O turnover (2024)

4.5 Crore

Retail F&O traders (up from 7L in 2019)

93%

Lost ₹1.81 lakh avg per person (FY22–FY24)

🔥 WHY THE BOOM HAPPENED

COVID lockdowns (2020)

Millions stuck at home, savings rising, markets crashed = perceived 'opportunity'. Zerodha, Upstox signups doubled.

Zero-fee platforms

₹0 account opening, ₹20/trade flat fee. F&O became accessible to anyone with a smartphone and Aadhaar.

YouTube & social media

Options trading 'gurus' promising consistent monthly income via strategies like Iron Condors to millions of followers.

Bull market confidence

Nifty from 7,500 to 18,000 (2020–2021). Easy money created false confidence. Leverage seemed risk-free.

Weekly expiry system

NSE introduced weekly option expiries (every Thursday). Short-term speculation exploded. Retail treated options as lottery tickets.

🛡️ SEBI RESPONSE + SIP LESSON

SEBI F&O Study (2023)

10 crore P&L data points. 89% lost money. Top 1% earned 51% of all profits. Access ≠ expertise.

SEBI reforms (2024)

Increased contract sizes, upfront premium collection, limited weekly expiries to 1 per exchange. Reduced speculative activity.

SIP investors comparison

₹26,000 crore/month in SIPs (2024). Disciplined, no-leverage, compound-return investors vs F&O gamblers. Night and day.

Teaching moment

Access to leverage is not the same as skill to use it. Lower costs and faster platforms increase participation — not quality of decisions.

Buffett comparison

'Risk comes from not knowing what you are doing.' 89% of F&O traders did not know what they were doing with leverage.



India Lesson: Lower transaction costs democratised access to leverage — but also democratised losses at scale.

What Theory Teaches Us vs. What Professionals Actually Do

Concept	Textbook Says	Professionals Do	Why It Differs
Margin Trading	Leverage increases expected return	Focus on survival before return. Capital preservation first.	Professionals know losing capital destroys future opportunities permanently
Maintenance Margin	Risk-control mechanism	Maintain excess collateral well above minimum thresholds	They prepare for unexpected volatility. Gaps can exceed protection margins.
Short Selling	Improves price discovery	Extensive research before shorting; size positions carefully	Timing risk is enormous. Unlimited upside = unlimited short seller risk.
Stop Losses	Set at maximum tolerable loss	Position sizing used FIRST; stops secondary	Margin calls will stop you out anyway — own risk size limits it before that.
Leverage Limits	Use maximum allowed for highest expected return	Target 1–2x max; 0x in concentrated single-name positions	One concentrated leveraged loss can wipe out years of gains. Survival matters.
Information Edge	More research = better returns	Information alone insufficient — timing and sizing critical	Being right too early with leverage = going broke before the idea plays out.



The core professional insight: You cannot compound returns if you do not survive. Leverage is the primary mechanism that makes investors non-survivors.

30 Thought-Provoking Discussion Questions

Leverage Logic	Margin Mechanics	Short Selling	Regulation & Risk	Behaviour & Future
Q1. Is leverage an investment tool or a gambling tool?	Q7. Who benefits most from margin trading?	Q13. Should short selling be banned during crises?	Q19. Should leverage be restricted for retail investors?	Q25. Why do smart investors repeatedly underestimate risk?
Q2. Why do sophisticated investors use leverage despite its dangers?	Q8. Should brokers have stricter margin requirements?	Q14. Does short selling destabilize markets?	Q20. Should regulators impose leverage limits?	Q26. Why do investors focus on gains rather than downside risk?
Q3. Does leverage magnify skill — or only luck?	Q9. Why do bull markets encourage excessive borrowing?	Q15. Are short sellers market villains or whistleblowers?	Q21. Is leverage more dangerous than derivatives?	Q27. Is leverage the primary cause of financial crises?
Q4. Is wealth creation impossible without some form of leverage?	Q10. Are margin calls necessary or harmful?	Q16. Why do governments restrict short selling?	Q22. Can AI reduce leverage-related mistakes?	Q28. Can AI reduce leverage-related mistakes?
Q5. Can leverage create false confidence?	Q11. Does forced selling worsen market crashes?	Q17. Did short sellers help uncover corporate frauds?	Q23. What lessons did Archegos teach markets?	Q29. Why do investors hate short sellers?
Q6. Does leverage increase market efficiency?	Q12. Should margin requirements vary by investor sophistication?	Q18. Is short selling ethical?	Q24. Can retail traders survive long-term using leverage?	Q30. Would you rather maximise return or survive?

Three Big Ideas

01

Leverage amplifies outcomes — it does not improve the quality of decisions

The mathematics are simple: a 20% stock move becomes a 100% gain or a 100% loss with 5x leverage. Leverage doesn't change probability — it changes consequences.

02

Margin calls mean the broker controls your position — not you

Maintenance margins exist to protect the broker, not the investor. When the call comes, timing belongs to the broker. The investor's analysis no longer matters — survival does.

03

Short selling is essential market discipline — but carries unlimited risk

Short sellers identified Enron, Wirecard, Adani governance issues before regulators. But timing is brutal. Unlimited upside risk means being right early = catastrophic losses.



Final Question: "Would you rather maximise return — or survive long enough to let compounding work?"

Debate : 'Retail investors should not be allowed to trade on margin.' → Institutional / Regulator / Retail / Broker groups